Reflection Groups and Coxeter Groups

Labix

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Contents

1 Root Systems

1.1 Abstract Root Systems and Simple Systems

Definition 1.1.1 (Abstract Root Systems)
Let E be a Euclidean space. Let $\Phi \subseteq E$ be a subset. We say that Φ is a root system if the following are true.

- Φ is finite and $0 \notin \Phi$.
- If $x \in \Phi$ and $\lambda x \in \Phi$, then $\lambda = \pm 1$.
- If $x, y \in \Phi$, then $S_x(y) \in \Phi$.

Note: Our definition of root systems does not include the requirement that Φ must span E, and that $2\frac{\langle x,y\rangle}{\langle y,y\rangle} \in \mathbb{Z}$.

Definition 1.1.2 (Simple Systems)
Let E be a Euclidean space. Let Φ be a root system. Let $\Delta \subseteq \Phi$ be a subset. We say that Δ is a simple system for Φ if the following are true.

- Δ is a basis for $\mathbb{R}\langle\Phi\rangle$.
- For any $r \in \Phi$, there is a decomposition
$$r = \sum_{b \in \Delta} k_b \cdot b$$
where $k_b \in \mathbb{R}$ are either all positive or all negative.

Lemma 1.1.3
Let E be a Euclidean space. Let Φ be a root system. Let Δ be a simple system of Φ. Let $x, y \in \Phi$ be distinct. Then we have
$$\langle x, y\rangle < 0$$

Lemma 1.1.4
Let E be a Euclidean space. Let Φ be a root system. Let Δ be a simple system of Φ. Then we have
$$G_\Phi = \langle S_x \mid x \in \Delta\rangle$$

Corollary 1.1.5
Let E be a Euclidean space. Let Φ be a root system. Let Δ be a simple system of Φ. Then for any $x \in \Phi$, there exists $y \in \Phi$ and $T \in G_\Phi$ such that $T(y) = x$.

1.2 Crystallographic Root Systems

Definition 1.2.1 (Crystallographic Root Systems)
Let E be a Euclidean space. Let Φ be a root system. We say that Φ is crystallographic if the following are true.

- Φ spans E.
- For any $x, y \in \Phi$, we have $2\frac{\langle x,y\rangle}{\langle y,y\rangle} \in \mathbb{Z}$.

Example 1.2.2
Let $\{e_1, \dots, e_{n+1}\}$ be the standard basis vectors of $\mathbb{R}^{n+1}$. Let E be the inner product space
$$E = \{(c_1, \dots, c_{n+1}) \subseteq \mathbb{R}^{n+1} \mid c_1 + \cdots + c_n = 0\} \cong \mathbb{R}^n$$
equipped with the standard inner product of $\mathbb{R}^{n+1}$. Then the set
$$R = \{e_i - e_j \mid 1 \leq i \neq j \leq n+1\}$$
is a root system of E.

Lemma 1.2.3
Let E be a finite dimensional inner product space over $\mathbb{R}$. Let R be a root system of E. Let $\alpha, \beta \in R$. Then the following are true.

- The value of $\langle\alpha,\beta\rangle\langle\beta,\alpha\rangle$ must lie in the set $\{0,1,2,3\}$
- The angle between α and β must lie in the set

$$\left\{\frac{\pi}{2},\frac{\pi}{3},\frac{2\pi}{3},\frac{\pi}{4},\frac{3\pi}{4},\frac{\pi}{6},\frac{5\pi}{6}\right\}$$

We can draw a table of the possible values of $\langle\alpha,\beta\rangle$ and the corresponding angle between α and β:

$\langle\alpha,\beta\rangle\langle\beta,\alpha\rangle$	$\langle\alpha,\beta\rangle$	$\langle\beta,\alpha\rangle$	θ
0	0	0	$\frac{\pi}{2}$
1	1	1	$\frac{\pi}{3}$
	-1	-1	$\frac{2\pi}{3}$
2	1	2	$\frac{\pi}{4}$
	-1	-2	$\frac{3\pi}{4}$
3	1	3	$\frac{\pi}{6}$
	-1	-3	$\frac{5\pi}{6}$

Here we assume $\|\alpha\| \leq \|\beta\|$ so we exclude for instance the case $\langle\alpha,\beta\rangle = 2$ and $\langle\beta,\alpha\rangle = 1$.

Example 1.2.4
The only root system in $\mathbb{R}^1$ is given by $R = \{\alpha, -\alpha\}$ for any $\alpha \in \mathbb{R}$.

Example 1.2.5
There are exactly four possible root systems in $\mathbb{R}^2$. Let $\alpha, \beta \in \mathbb{R}^2$ be linearly independent such that $\|\alpha\| \leq \|\beta\|$. Let θ be the angle between α and β.

- $A_1 \times A_1 = \{\pm\alpha, \pm\beta\}$ and $\theta = \frac{\pi}{2}$.
- $A_2 = \{\pm\alpha, \pm\beta, \pm(\alpha+\beta)\}$ and $\theta = \frac{2\pi}{3}$.
- $B_2 = \{\pm\alpha, \pm\beta, \pm(\alpha+\beta), \pm(2\alpha+\beta)\}$ and $\theta = \frac{3\pi}{4}$.
- $G_2 = \{\pm\alpha, \pm\beta, \pm(\alpha+\beta), \pm(2\alpha+\beta), \pm(3\alpha+\beta), \pm(3\alpha+2\beta)\}$

Definition 1.2.6 (Base of a Crystallographic Root System)
Let E be a Euclidean space. Let Φ be a root system. Let $B \subseteq \Phi$ be a subset. We say that B is a base for Φ if the following are true.

- B is a basis for $\mathbb{R}\langle\Phi\rangle = E$.
- For any $r \in \Phi$, there is a decomposition

$$r = \sum_{b \in \Delta} k_b \cdot b$$

where $k_b \in \mathbb{Z}$ are either all positive or all negative.

Proposition 1.2.7
Let E be a finite dimensional inner product space over $\mathbb{R}$. Let Φ be a crystallographic root system of E. Then Φ admits a base.

1.3 Decomposing Root Systems into Irreducibles

Definition 1.3.1 (Irreducible Root Systems)
Let $(E, (-,-))$ be a finite dimensional inner product space over $\mathbb{R}$. Let R be a crystallographic root system of E. We say that R is reducible if there exists two disjoint crystallographic root systems R_1 and R_2 of E such that $(r_1, r_2) = 0$ for $r_1 \in R_1$ and $r_2 \in R_2$. Otherwise, we say that R is irreducible.

Proposition 1.3.2
Let E be a finite dimensional inner product space over $\mathbb{R}$. Let R be a crystallographic root system of E. Then the following are true.

- There is a decomposition
$$R = R_1 \amalg \cdots \amalg R_k$$
where each R_i is an irreducible crystallographic root system of $E_i = \mathbb{R}\langle R_i \rangle$
- There is a decomposition
$$E = E_1 \oplus \cdots \oplus E_k$$
where E_i is orthogonal to E_j for $i \neq j$.

Definition 1.3.3 (Isomorphic Root Systems)
Let E, F be a finite dimensional inner product spaces over $\mathbb{R}$. Let R and S be crystallographic root systems of E and F respectively. We say that R and S are isomorphic if there exists an isomorphism
$$\phi : E \to F$$
of vector spaces such that the following are true.

- $\phi(R) = S$
- For all $a, b \in R$, $\langle \phi(a), \phi(b) \rangle = \langle a, b \rangle$.

Example 1.3.4
The following are true.

- $A_1 \times A_1 = \{\pm\alpha, \pm\beta\}$ is reducible into the product of A_1 and A_1.
- A_2, B_2, G_2 are irreducible.

1.4 The Weyl Group of a Root System

Definition 1.4.1 (Weyl Group of a Root System)
Let E be a Euclidean space. Let Φ be a crystallographic root system. Define the Weyl group of Φ to be
$$W(\Phi) = G_\Phi = \langle S_x \mid x \in \Phi \rangle$$

2 Dynkin Diagrams

2.1 Dynkin Diagrams and Cartan Matrices

Definition 2.1.1 (The Dynkin Diagram of a Root System)
Let E be a finite dimensional inner product space over $\mathbb{R}$. Let R be a crystallographic root system of E. Let B be a base of R. Define a graph $\Delta(R)$ as follows.

- There is one vertex v_b for each $b \in B$
- For any two vertices v_a and v_b, there are $d_{a,b} = \langle a, b\rangle\langle b, a\rangle$ number of undirected edges between v_a and v_b.

Proposition 2.1.2
Let E, F be a finite dimensional inner product spaces over $\mathbb{R}$. Let R and S be crystallographic root systems of E and F respectively. Then $R \cong S$ if and only if $\Delta(R) \cong \Delta(S)$.

Proposition 2.1.3
Let E be a finite dimensional inner product space. Let R be a crystallographic root system of E. Then R is irreducible if and only if $\Delta(R)$ is a connected graph.

Definition 2.1.4 (The Cartan Matrix)
Let E be a finite dimensional inner product space over $\mathbb{R}$. Let R be a crystallographic root system of E. Let $B = \{\alpha_1, \dots, \alpha_r\}$ be a base of R. Define the Cartan matrix of R to be

$$\begin{pmatrix} \langle \alpha_1, \alpha_1\rangle & \cdots & \langle \alpha_1, \alpha_r\rangle \\ \vdots & \ddots & \vdots \\ \langle \alpha_r, \alpha_1\rangle & \cdots & \langle \alpha_r, \alpha_r\rangle \end{pmatrix}$$

Proposition 2.1.5
Let E be a finite dimensional inner product space over $\mathbb{R}$. Let R be a crystallographic root system of E. Then the Cartan matrix of R is unique up to reordering.

2.2 Classification of Irreducible Root Systems

Theorem 2.2.1 (Classification of Irreducible Root Systems)
Let E be a finite dimensional inner product space over $\mathbb{R}$. Let R be an irreducible crystallographic root system of E. Then $\Delta(R)$ is isomorphic to one of the following:

- A_n for $n \geq 1$.
- B_n for $n \geq 2$.
- C_n for $n \geq 3$.
- D_n for $n \geq 4$.
- G_2, F_4, E_6, E_7, E_8.

Conversely, any of the above types can be realized as the Dynkin diagram of some irreducible root system.

3 Reflection Groups

3.1 Reflections

Definition 3.1.1 (Reflections)
Let E be a Euclidean space. Let $T : E \to E$ be a linear map. We say that T is a reflection if

$$\{v \in E \mid T(v) = v\}$$

is an $(n-1)$-dimensional subspace of E.

Definition 3.1.2 (The Reflection Associated to a Vector)
Let E be a Euclidean space. Let $x \in E$. Define the reflection associated to E to be the map $S_x : E \to E$ given by

$$S_x(z) = z - 2\frac{\langle x, z\rangle}{\langle x, x\rangle}x$$

Lemma 3.1.3
Let E be a Euclidean space. Let $x \in E$. Then the following are true.
- S_x is a reflection.
- $S_x = S_{\lambda x}$ for any $\lambda \in \mathbb{R} \setminus \{0\}$.

Definition 3.1.4 (Root of a Reflection)
Let E be a Euclidean space. Let $T : E \to E$ be a reflection. Let $x \in E$. We say that x is a root of T if we have $T = S_x$.

Proposition 3.1.5
Let E be a Euclidean space. Let $T : E \to E$ be a reflection. Then the following are true.
- T admits a root.
- The set of roots of T is given by

$$\{v \in E \mid T(v) = v\}^\perp$$

Lemma 3.1.6
Let E be a Euclidean space. Let $T : E \to E$ be a reflection. Then $T^2 = \text{id}_E$.

Definition 3.1.7 (Finite Reflection Groups)
Let E be a Euclidean space. Let $G < O(E)$. We say that G is a finite reflection group if the following are true.
- G is finite.
- G is generated by reflections.

Example 3.1.8
Consider the representation

$$D_{2n} \cong \left\langle \begin{pmatrix} \cos(2\pi/n) & -\sin(2\pi/n) \\ \sin(2\pi/n) & \cos(2\pi/n) \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \right\rangle \leq O(\mathbb{R}^2)$$

Then D_{2n} is a finite reflection group.

Example 3.1.9
Consider S_n as a subgroup of $O(\mathbb{R}^n)$ via the injection

$$\sigma \mapsto (e_i \mapsto e_{\sigma(i)})$$

Then S_n is a finite reflection group.

Proof
Indeed, S_n is generated by transpositions, and transpositions are reflections since for a transposition $\begin{pmatrix} i & j \end{pmatrix}$, the corresponding element in $O(\mathbb{R}^n)$ fixes e_k for $k \neq i, j$ and transposes $e_i - e_j$ with $e_j - e_i$. ■

Definition 3.1.10 (Root Systems of a Finite Reflection Group)
Let E be a Euclidean space. Let G be a finite reflection group of E. Define the root system of G to be
$$\Phi_G = \{x \in E \mid S_x \in G \text{ and } \|x\| = 1\}$$

Lemma 3.1.11
Let E be a Euclidean space. Let G be a finite reflection group of E. Then Φ_G is a root system.

3.2 The Reflection Group of a Root System

Definition 3.2.1 (The Finite Reflection Group of a Root System)
Let E be a Euclidean space. Let $\Phi \subseteq E$ be a root system. Define the finite reflection group associated to Φ to be
$$G_\Phi = \langle S_x \mid x \in \Phi \rangle$$

Lemma 3.2.2
Let E be a Euclidean space. Let $\Phi \subseteq E$ be a root system. Then G_Φ is a finite reflection group.

Note: There can be multiple different root systems that give rise to the same finite reflection group. Note: $\{S_x \mid x \in \Phi_G\}$ might not be the complete set of reflections of G.

4 Coxeter Groups

4.1 Coxeter Groups

Definition 4.1.1 (Coxeter Groups)
Let G be a group. We say that G is a Coxeter group if G admits a presentation

$$G \cong \langle r_1, \dots, r_n \mid (r_i r_j)^{m_{ij}} \rangle$$

where $m_{ii} = 1$ and $m_{ij} \geq 2$ or ∞ for $i \neq j$.

Proposition 4.1.2
Let E be a Euclidean space. Let G be a finite reflection group of E. Then G is a Coxeter group.